

Stochastic Processes

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1 Introduction

1.1 Probability

Definition 1.1. (*Probability Space*)

A probability space is a triple $(\Omega, \mathcal{F}, P)$. The sample space Ω is the set of outcomes. The event field $\mathcal{F}$ is a σ -field of subsets of Ω :

- $\Omega \in \mathcal{F}$;
- $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$;
- $A_k \in \mathcal{F}$ implies $\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}$.

Definition 1.2. (*Probability*)

Given a probability space $(\Omega, \mathcal{F}, P)$, a set function $P : \mathcal{F} \rightarrow \mathbb{R}$ is a probability if

- $P(A) \geq 0$ for every $A \in \mathcal{F}$;
- $P(\Omega) = 1$;
- whenever $A_i \in \mathcal{F}$ are pairwise disjoint, $P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$.

Definition 1.3. (*Random Variable*)

Given a probability space $(\Omega, \mathcal{F}, P)$, a map $X : \Omega \rightarrow \mathbb{R}$ is a random variable if $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$.

Definition 1.4. (*Distribution Function*)

Given a random variable X , the distribution function is $F_X(x) = P(X \leq x)$.

Definition 1.5. (*Expectation*)

If the integral exists,

$$E(X) = \int_{-\infty}^{+\infty} x \, dF_X(x).$$

This Riemann-Stieltjes integral is $\sum_i x_i (F_X(x_i) - F_X(x_i -))$ in the discrete case and $\int x f_X(x) \, dx$ in the continuous case. For a measurable g ,

$$E(g(X)) = \int g(x) \, dF_X(x).$$

Proposition 1.1.

Expectation is linear whether or not the summands are independent.

Definition 1.6. (*Variance, Covariance, and Correlation*)

The variance, covariance, and correlation are

$$\text{Var}(X) = E(X - EX)^2 = E(X^2) - (EX)^2,$$

$$\text{Cov}(X, Y) = E((X - EX)(Y - EY)) = E(XY) - E(X)E(Y),$$

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

Proposition 1.2.

$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}$, and therefore $|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X) \text{Var}(Y)}$.

1.2 Conditional Expectation

Definition 1.7. (*Conditional Expectation*)

Let $E|X| < \infty$. A random variable $E(X|Y)$ is a conditional expectation of X given Y if $E(X|Y) = g(Y)$ for some measurable g with $E|g(Y)| < \infty$, and

$$E(g(Y)I_D(Y)) = E(XI_D(Y))$$

for every Borel set D .

Proposition 1.3.

- $E(E(X|Y)) = E(X)$.
- If $h(Y)$ is bounded, then $E(g(X)h(Y) | Y) = h(Y)E(g(X)|Y)$. In particular $E(E(X|Y)h(Y)) = E(Xh(Y))$.
- If X and Y are independent, then $E(X|Y) = E(X)$.

Remark.

The proposition says that $X - E(X|Y)$ is orthogonal to every bounded function of Y : $\text{Cov}(X - E(X|Y), h(Y)) = 0$ whenever the covariance exists.

Theorem 1.1. (Least Squares)

If $E(X^2) < \infty$ and $g(Y)$ is square-integrable, then

$$E(X - g(Y))^2 \geq E(X - E(X|Y))^2.$$

Proof.

Write $X - g(Y) = (X - E(X|Y)) + (E(X|Y) - g(Y))$. In the expanded square, the cross term vanishes because $E(X|Y) - g(Y)$ is a function of Y and is orthogonal to $X - E(X|Y)$. What remains is

$$E(X - E(X|Y))^2 + E(E(X|Y) - g(Y))^2,$$

and the second summand is nonnegative. ■

Theorem 1.2. (Independence)

If X and Y are independent and $E|h(X, Y)| < \infty$, then $E(h(X, Y)|Y) = g(Y)$, where $g(y) = E(h(X, y))$.

Proof.

Independence gives $dF_{X,Y} = dF_X dF_Y$. For every Borel set D ,

$$E(h(X, Y)I_D(Y)) = \int \int h(x, y)I_D(y) dF_X(x) dF_Y(y) = E(g(Y)I_D(Y)).$$

■

Theorem 1.3. (Tower Property)

If $1 \leq m \leq n$ and $E|X| < \infty$, then

$$E(X | Y_1, \dots, Y_m) = E(E(X | Y_1, \dots, Y_n) | Y_1, \dots, Y_m).$$

Proof.

It is enough to show that $X - E(X|Y_1, \dots, Y_n)$ has conditional expectation zero given $(Y_1, \dots, Y_m)$. For every bounded measurable $h(y_1, \dots, y_m)$, the product $h(Y_1, \dots, Y_m)$ is a bounded function of $(Y_1, \dots, Y_n)$, so

$$E(Xh(Y_1, \dots, Y_m)) = E(E(X|Y_1, \dots, Y_n)h(Y_1, \dots, Y_m)).$$

The two sides therefore agree, and the difference $X - E(X|Y_1, \dots, Y_n)$ is orthogonal to every bounded function of $(Y_1, \dots, Y_m)$. ■

Definition 1.8. (*Conditional Variance*)

$$\text{Var}(Y|X) = E((Y - E(Y|X))^2 | X).$$

Theorem 1.4. (*Law of Total Variance*)

If $E(Y^2) < \infty$, then

$$\text{Var}(Y) = E(\text{Var}(Y|X)) + \text{Var}(E(Y|X)).$$

In particular $\text{Var}(Y) \geq E(\text{Var}(Y|X))$, $E(\text{Var}(Y|X_1)) \geq E(\text{Var}(Y|X_1, X_2))$.

Proof.

$$\text{Var}(Y) = E(Y^2) - (EY)^2 = E(E(Y^2|X)) - (E(E(Y|X)))^2.$$

Substitute $E(Y^2|X) = \text{Var}(Y|X) + (E(Y|X))^2$. The same decomposition given X_1 , applied to the conditioning σ -field of (X_1, X_2) , yields

$$\text{Var}(Y|X_1) = E(\text{Var}(Y|X_1, X_2)|X_1) + \text{Var}(E(Y|X_1, X_2)|X_1).$$

The second summand is nonnegative. Taking expectations gives the comparison. ■

Theorem 1.5. (*Dominated Convergence*)

Let $EY < \infty$ with $Y \geq 0$, and suppose $|X_n| \leq Y$ almost surely. If $X_n \xrightarrow{a.s.} X$, then X is integrable and

$$E(X_n) \xrightarrow{a.s.} E(X).$$

If instead $X_n \xrightarrow{P} X$, the same convergence holds in probability.

1.3 Stochastic Process

Definition 1.9. (Stochastic Process)

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $\mathbb{T}$ be a time set. A stochastic process is a family $X = \{X_t : t \in \mathbb{T}\}$ such that X_t is a random variable for every $t \in \mathbb{T}$.

Definition 1.10. (Finite-Dimensional Distributions)

For $t_1 < \dots < t_n$ in $\mathbb{T}$, the distribution function

$$F_{t_1, \dots, t_n}(x_1, \dots, x_n) = P(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n)$$

is a finite-dimensional distribution of X . Every process has one, and it is unique.

Proposition 1.4. (Symmetry and Consistency)

The finite-dimensional distribution of a process satisfy

- symmetry: for every permutation $(j_1, \dots, j_n)$ of $(1, \dots, n)$,

$$F_{t_{j_1}, \dots, t_{j_n}}(x_{j_1}, \dots, x_{j_n}) = F_{t_1, \dots, t_n}(x_1, \dots, x_n);$$

- consistency: if $m < n$, then

$$F_{t_1, \dots, t_n}(x_1, \dots, x_m, +\infty, \dots, +\infty) = F_{t_1, \dots, t_m}(x_1, \dots, x_m).$$

Definition 1.11. (Mean, Covariance, and Correlation Functions)

For a real process $\{X_t : t \in \mathbb{T}\}$,

$$m(t) = E(X_t), \quad \text{Var}(X_t) = E(X_t - m(t))^2,$$

$$R(s, t) = E(X_s X_t), \quad C_X(s, t) = \text{Cov}(X_s, X_t) = R(s, t) - m(s)m(t).$$

The correlation function of the process is R , and the correlation coefficient is

$$\rho(s, t) = \frac{C_X(s, t)}{\sqrt{\text{Var}(X_s) \text{Var}(X_t)}}.$$

2 Poisson Process

2.1 Homogeneous Poisson Process

Definition 2.1. (Counting Process)

A counting process $\{N_t : t \geq 0\}$ satisfies $N_0 = 0$, takes values in $\{0, 1, 2, \dots\}$, and

has nondecreasing paths. Then N_t counts events in $[0, t]$, and $N_t - N_s$ counts events in $(s, t]$.

Definition 2.2. (*Independent and Stationary Increments*)

A process X has independent increments if, for every $t_0 < t_1 < \dots < t_n$, the increments $X_{t_1} - X_{t_0}, \dots, X_{t_n} - X_{t_{n-1}}$ are independent. It has stationary increments if, for $0 \leq s < t$, the increment $X_t - X_s$ has the same distribution as $X_{t-s} - X_0$.

Definition 2.3. (*Homogeneous Poisson Process*)

A counting process $\{N_t : t \geq 0\}$ is a homogeneous Poisson process of intensity $\lambda > 0$ if

- it has independent increments and stationary increments;
- for every $t > 0$ and every sufficiently small $h > 0$,

$$P(N_{t+h} - N_t = 1) = \lambda h + o(h),$$

$$P(N_{t+h} - N_t = 0) = 1 - \lambda h + o(h).$$

Theorem 2.1.

If $\{N_t : t \geq 0\}$ is a homogeneous Poisson process of intensity λ , then for every $s \geq 0$ and $t > 0$,

$$N_{s+t} - N_s \sim \text{Poisson}(\lambda t).$$

In particular, $P(N_t = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}$ for $k = 0, 1, 2, \dots$

Proof.

Write $p_k(t) = P(N_t = k)$. Independent and stationary increments give, for $h > 0$,

$$p_0(t+h) = p_0(t)(1 - \lambda h + o(h)).$$

Dividing by h and letting $h \rightarrow 0$ yields $p'_0(t) = -\lambda p_0(t)$. Since $N_0 = 0$, one has $p_0(0) = 1$, so $p_0(t) = e^{-\lambda t}$.

For $k \geq 1$ the same splitting, and the estimate $P(\text{increment} \geq 2) = o(h)$, gives

$$p_k(t+h) = p_k(t)(1 - \lambda h + o(h)) + p_{k-1}(t)(\lambda h + o(h)) + o(h),$$

hence $p'_k(t) = -\lambda p_k(t) + \lambda p_{k-1}(t)$.

The probability generating function $\Phi(t, z) = \sum_{k=0}^{\infty} p_k(t) z^k$ satisfies $\Phi(0, z) = 1$ and

$$\partial_t \Phi(t, z) = -\lambda(1 - z)\Phi(t, z).$$

Therefore $\Phi(t, z) = \exp(-\lambda t(1 - z)) = e^{-\lambda t} \sum_{k=0}^{\infty} (\lambda t)^k \frac{z^k}{k!}$, and the coefficient of z^k is the Poisson probability. ■

Definition 2.4. (*Second Definition*)

A counting process $\{N_t : t \geq 0\}$ is a homogeneous Poisson process of intensity λ if $N_0 = 0$, the process has independent increments, and $N_{s+t} - N_s \sim \text{Poisson}(\lambda t)$ for all $s, t \geq 0$.

Proposition 2.1.

If $\{N_t\}$ is a homogeneous Poisson process of intensity λ , then $E(N_t) = \text{Var}(N_t) = \lambda t$ and

$$\text{Cov}(N_s, N_t) = \lambda \min(s, t).$$

Proof.

The mean and variance are those of $\text{Poisson}(\lambda t)$. If $s \leq t$, independent increments give

$$\text{Cov}(N_s, N_t) = \text{Cov}(N_s, N_s + (N_t - N_s)) = \text{Var}(N_s) = \lambda s.$$

■

Theorem 2.2.

If $N^{(1)}, \dots, N^{(m)}$ are independent homogeneous Poisson processes of intensities $\lambda_1, \dots, \lambda_m$, then $N_t = N_t^{(1)} + \dots + N_t^{(m)}$ is a homogeneous Poisson process of intensity $\lambda_1 + \dots + \lambda_m$.

Theorem 2.3.

Let $\{N_t\}$ be a homogeneous Poisson process of intensity λ . Mark each event independently as type 1 with probability p and type 2 with probability $1 - p$, and let $N_t^{(1)}$ and $N_t^{(2)}$ be the counts of the two types in $[0, t]$. Then for each fixed t ,

$$N_t^{(1)} \sim \text{Poisson}(\lambda p t), \quad N_t^{(2)} \sim \text{Poisson}(\lambda(1 - p)t),$$

and $N_t^{(1)}$ is independent of $N_t^{(2)}$.

Proof.

Condition on the total count and use the binomial thinning:

$$P\left(N_t^{(1)} = i, N_t^{(2)} = j\right) = e^{-\lambda t} \frac{(\lambda t)^{i+j}}{(i+j)!} \cdot \binom{i+j}{i} p^i (1-p)^j,$$

which factors as the product of the two Poisson probabilities. ■

2.2 Arrival Time

Definition 2.5. (Arrival and Interarrival Time)

Let $\tau_0 = 0$ and let τ_n be the time of the n -th event. The interarrival time is $X_n = \tau_n - \tau_{n-1}$ for $n \geq 1$.

Theorem 2.4.

If $\{N_t\}$ is a homogeneous Poisson process of intensity λ , then $\tau_n \sim \Gamma(n, \lambda)$:

$$f_{\tau_n}(t) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}, \quad t \geq 0.$$

Proof.

$$P(\tau_n \leq t) = P(N_t \geq n) = \sum_{k=n}^{\infty} e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

Differentiating term by term,

$$\frac{d}{dt} \left(e^{-\lambda t} \frac{(\lambda t)^k}{k!} \right) = -\lambda e^{-\lambda t} \frac{(\lambda t)^k}{k!} + \lambda e^{-\lambda t} \frac{(\lambda t)^{k-1}}{(k-1)!}.$$

Summing from $k = n$ to ∞ and reindexing the second sum yields the density. ■

Theorem 2.5.

On $0 < t_1 < \dots < t_n < \infty$, the joint density of $(\tau_1, \dots, \tau_n)$ is $\lambda^n e^{-\lambda t_n}$.

Proof.

If $h > 0$ is small enough that the windows $(t_i - \frac{h}{2}, t_i + \frac{h}{2})$ are disjoint, the probability of one event in each window and none elsewhere on $[0, t_n + \frac{h}{2}]$ equals $(\lambda h)^n e^{-\lambda(t_n + \frac{h}{2})}$. Dividing by h^n and letting $h \rightarrow 0^+$ yields the density $\lambda^n e^{-\lambda t_n}$. ■

Theorem 2.6.

A counting process is a homogeneous Poisson process of intensity λ if and only if its interarrival times $X_i \sim \text{Exp}(\lambda)$ i.i.d..

Proof.

Suppose N is Poisson of intensity λ . The map $(x_1, \dots, x_n) \mapsto (x_1, x_1 + x_2, \dots, x_1 + \dots + x_n)$ has Jacobian determinant 1. Substituting $t_n = x_1 + \dots + x_n$ into the joint density of the arrival times gives, for $x_i > 0$,

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \lambda^n e^{-\lambda(x_1 + \dots + x_n)} = \prod_{i=1}^n \lambda e^{-\lambda x_i}.$$

Thus the interarrival times are i.i.d. exponential of rate λ .

Conversely, start from i.i.d. exponential interarrival times, set $\tau_n = X_1 + \dots + X_n$ and $N_t = \sup\{n : \tau_n \leq t\}$. Then $\tau_n \sim \Gamma(n, \lambda)$, so

$$P(N_t = n) = P(\tau_n \leq t) - P(\tau_{n+1} \leq t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

The memoryless property of the exponential law makes the time until the next event, after any fixed time, exponential of rate λ and independent of the history. ■

Theorem 2.7.

If $\{N_t\}$ is a homogeneous Poisson process of intensity λ and $n \geq 1$, then conditionally on $N_t = n$ the arrival times $(\tau_1, \dots, \tau_n)$ have density

$$f(t_1, \dots, t_n \mid N_t = n) = \frac{n!}{t^n}, \quad 0 < t_1 < \dots < t_n < t.$$

Namely, they are distributed as the order statistics of n i.i.d. uniform random variables on $[0, t]$.

Proof.

For the joint density, take disjoint windows of width h about $t_1 < \dots < t_n$. The probability of one event in each window and none elsewhere in $[0, t]$ is $e^{-\lambda t} (\lambda h)^n$, and $P(N_t = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$. The ratio is $n! \left(\frac{h}{t}\right)^n$. Dividing by h^n produces $\frac{n!}{t^n}$. ■

Example.

Passengers arrive at a stop as a homogeneous Poisson process of intensity λ per minute. A bus clears the queue every 10 minutes. Let τ_k be the arrival time of the k -th passenger during a ten-minute cycle, and let $X = \sum_{k=1}^{N_{10}} (10 - \tau_k)$ be the total waiting time in that cycle. Then

$$E(X) = E(E(X \mid N_{10})) = E(5N_{10}) = 50\lambda.$$

2.3 Nonhomogeneous Poisson Process

Definition 2.6. (*Nonhomogeneous Poisson Process*)

Let $\lambda(t) > 0$ be locally integrable. A counting process $\{N_t : t \geq 0\}$ is a nonhomogeneous Poisson process of intensity $\lambda(t)$ if it has independent increments and for every $t \geq 0$ and every sufficiently small $h > 0$,

$$P(N_{t+h} - N_t = 1) = \lambda(t)h + o(h),$$

$$P(N_{t+h} - N_t = 0) = 1 - \lambda(t)h + o(h).$$

Theorem 2.8.

A counting process is a nonhomogeneous Poisson process of intensity $\lambda(t)$ if and only if it has independent increments and

$$N_t - N_s \sim \text{Poisson}(m(t) - m(s)), \quad 0 \leq s < t,$$

where $m(t) = \int_0^t \lambda(u) du$.